

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:4

Total Marks:40

1. Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.
2. Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.
3. Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.
4. Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

Code of Conduct:

I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

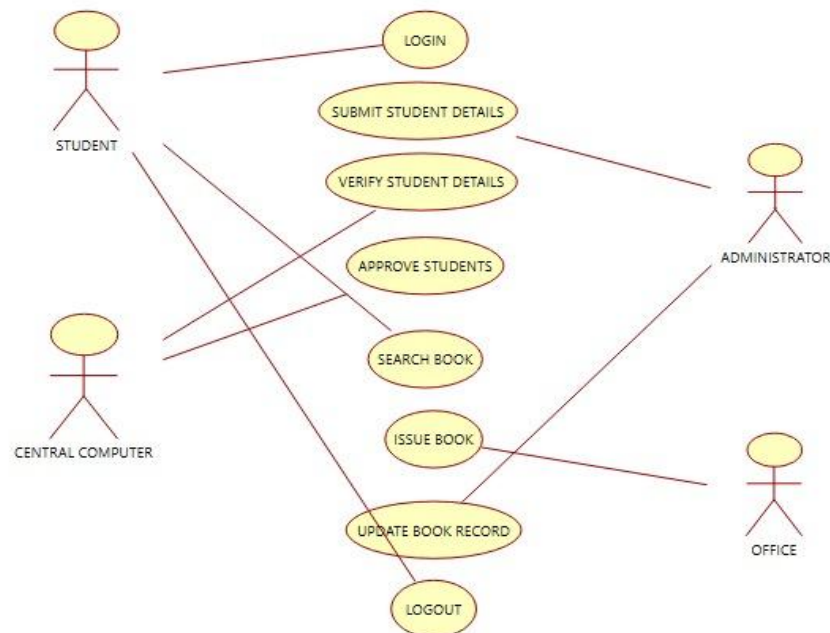
QUESTION 1: Book Bank System

Aim:

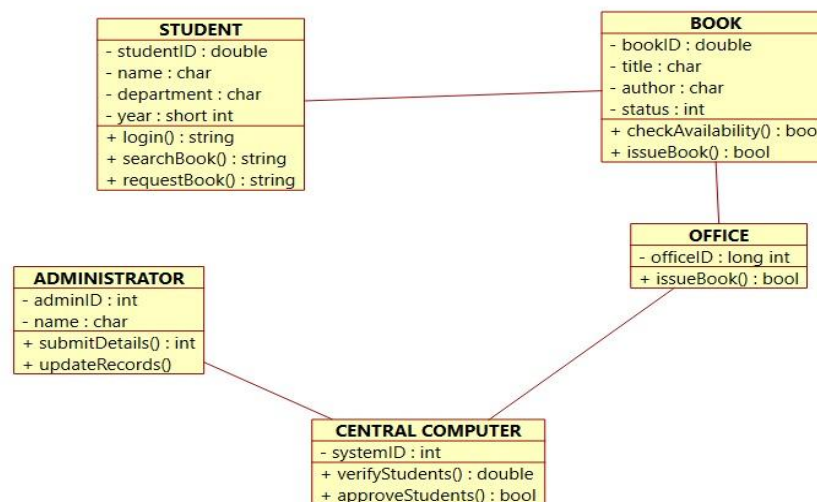
To model the Book Bank System using UML diagrams in Umbrello for verifying student details and issuing books efficiently.

Procedure:

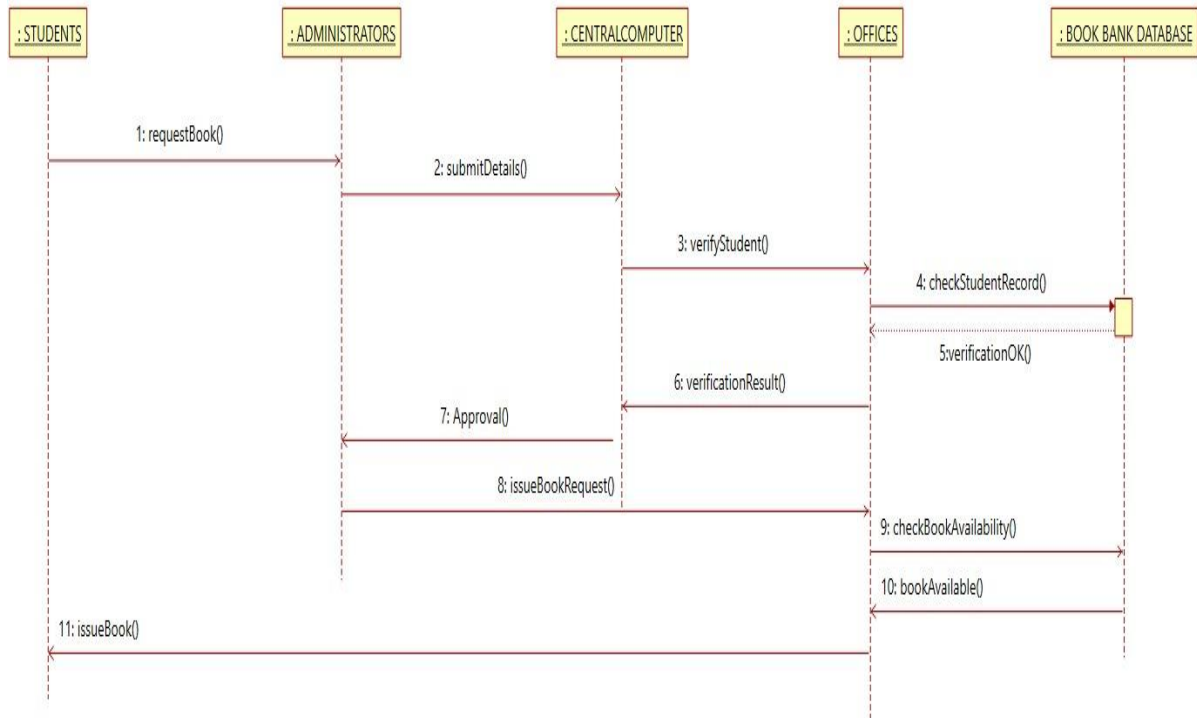
Use Case Diagram



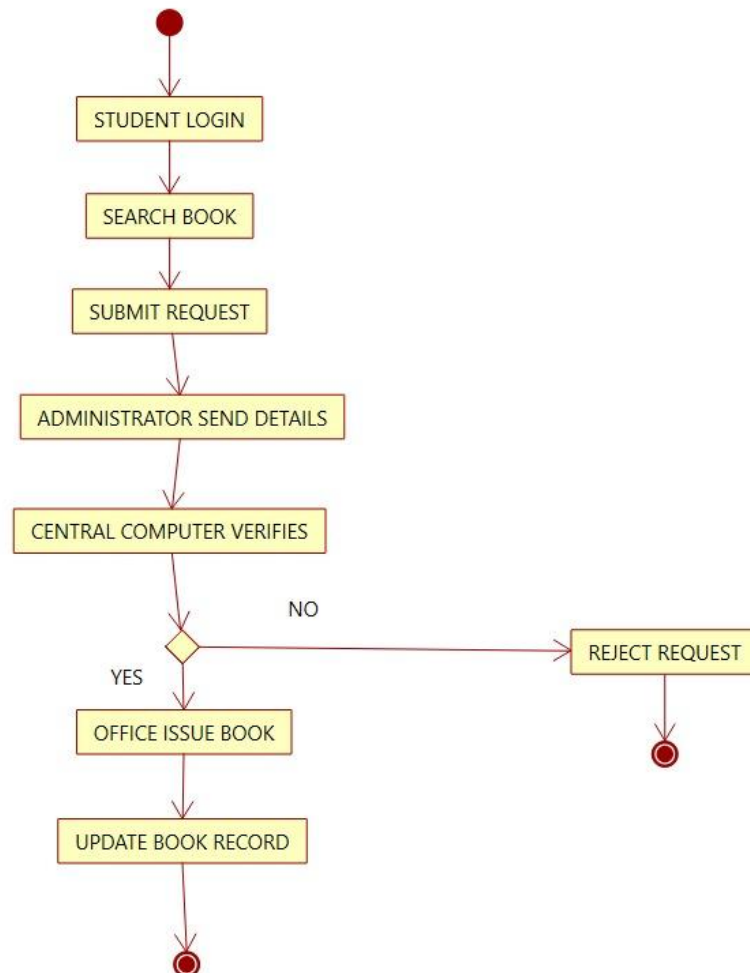
Class Diagram



Sequence Diagram



Activity Diagram



Result:

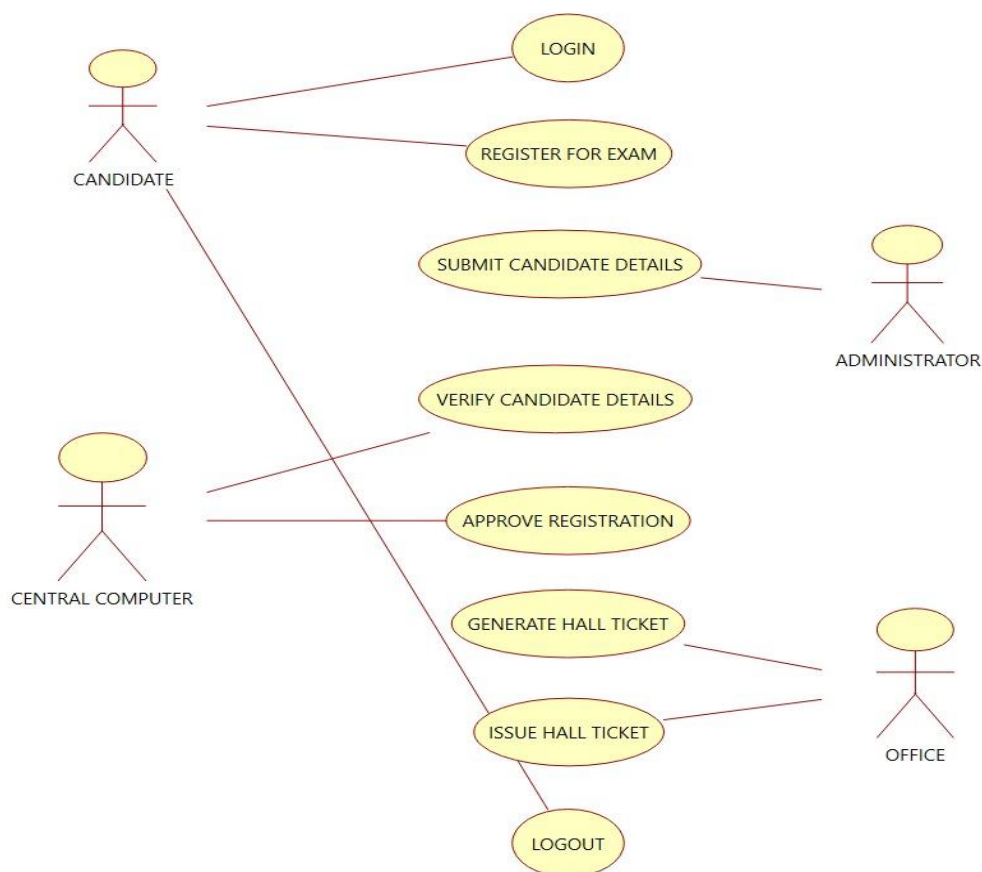
The Book Bank System was successfully modeled using Umbrello UML with Use Case, Class, Sequence, and Activity diagrams. The system verifies student details through the central computer and issues books through the office.

Analysis:

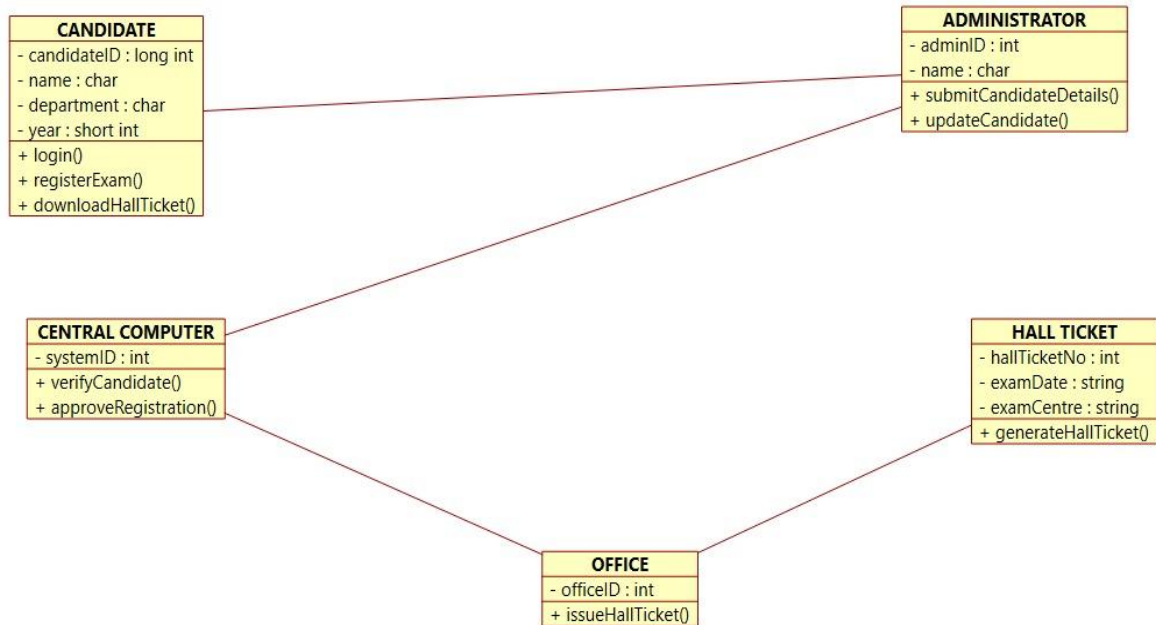
The UML model clearly represents the interactions among the Student, Administrator, Central Computer, Office, and Book entities. It demonstrates the complete workflow from student verification to book issuance, ensuring accurate system analysis and object-oriented design.

QUESTION 2: Exam Registration Portal**Aim:**

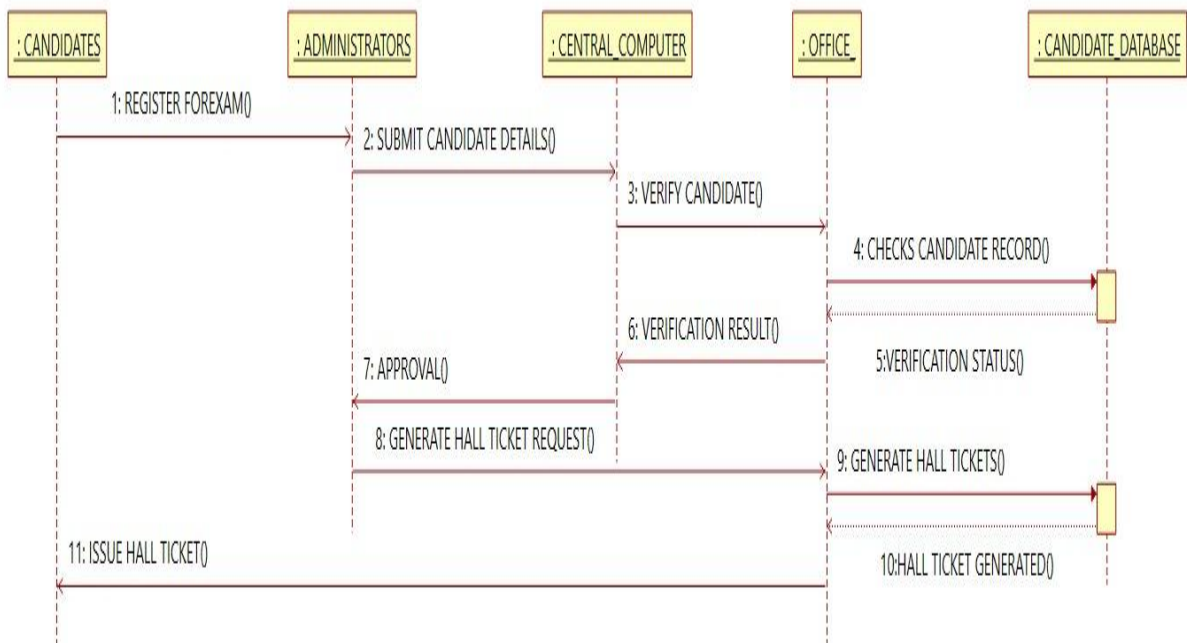
To develop a UML model for the Exam Registration Portal using Umbrello UML Modeller, showing the process of candidate verification and hall ticket generation.

Procedure:**Use Case Diagram**

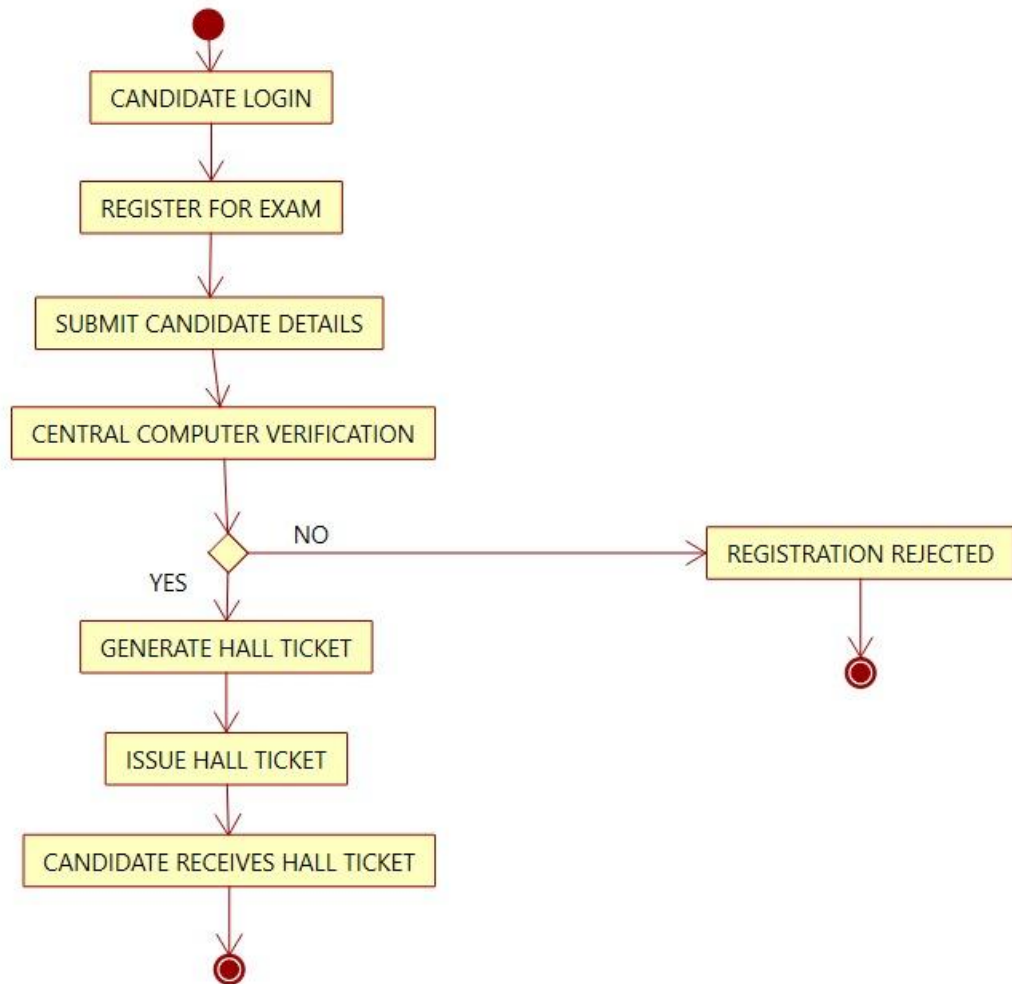
Class Diagram



Sequence Diagram



Activity Diagram



Result:

The Exam Registration Portal was successfully modeled using Umbrello UML Modeller. The UML diagrams represent candidate registration, verification by the Central Computer, approval, hall ticket generation, and issuance.

Analysis:

The UML model clearly represents the interaction between the Candidate, Administrator, Central Computer, and Office. It demonstrates the complete workflow from exam registration through candidate verification to hall ticket issuance, ensuring proper object-oriented system analysis and design.

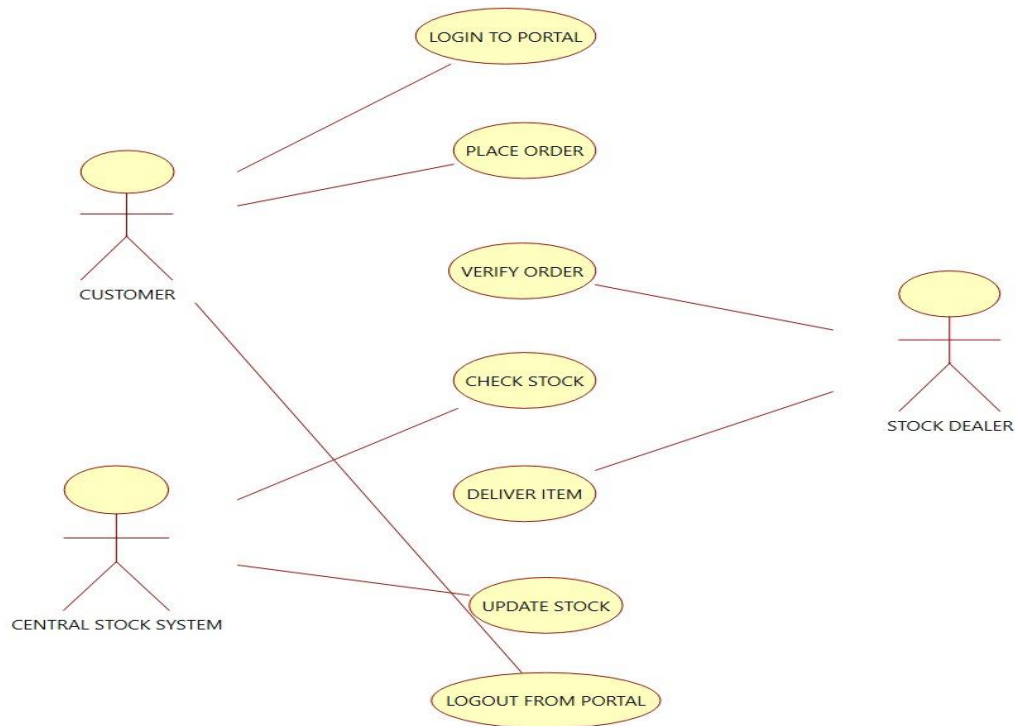
QUESTION 3: Stock Maintenance System

Aim:

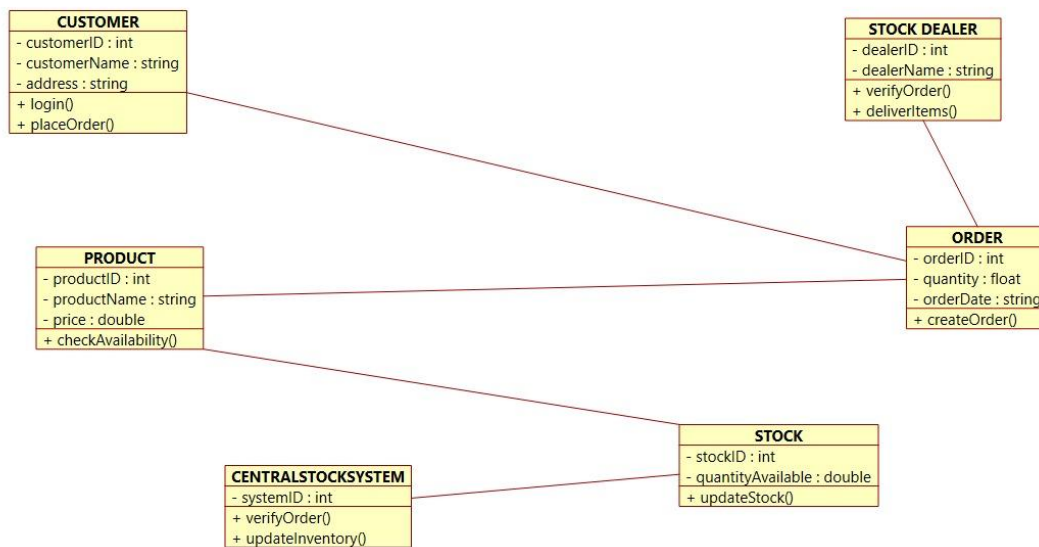
To develop a UML model for the Stock Maintenance System using Umbrello UML Modeller to represent the ordering, verification, delivery, and stock update process.

Procedure:

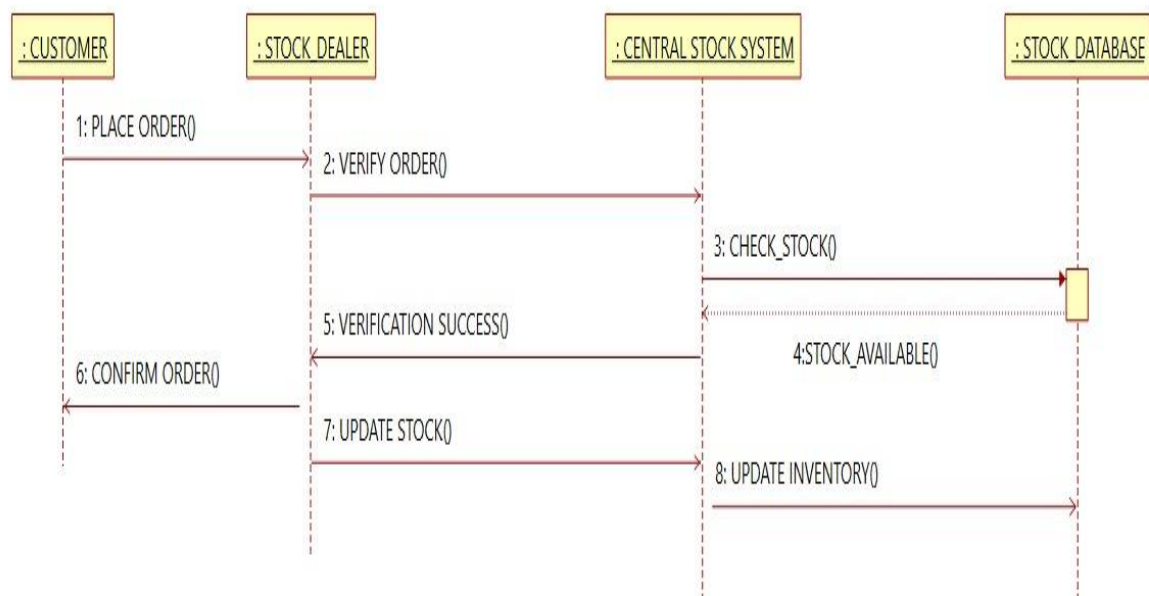
Use Case Diagram



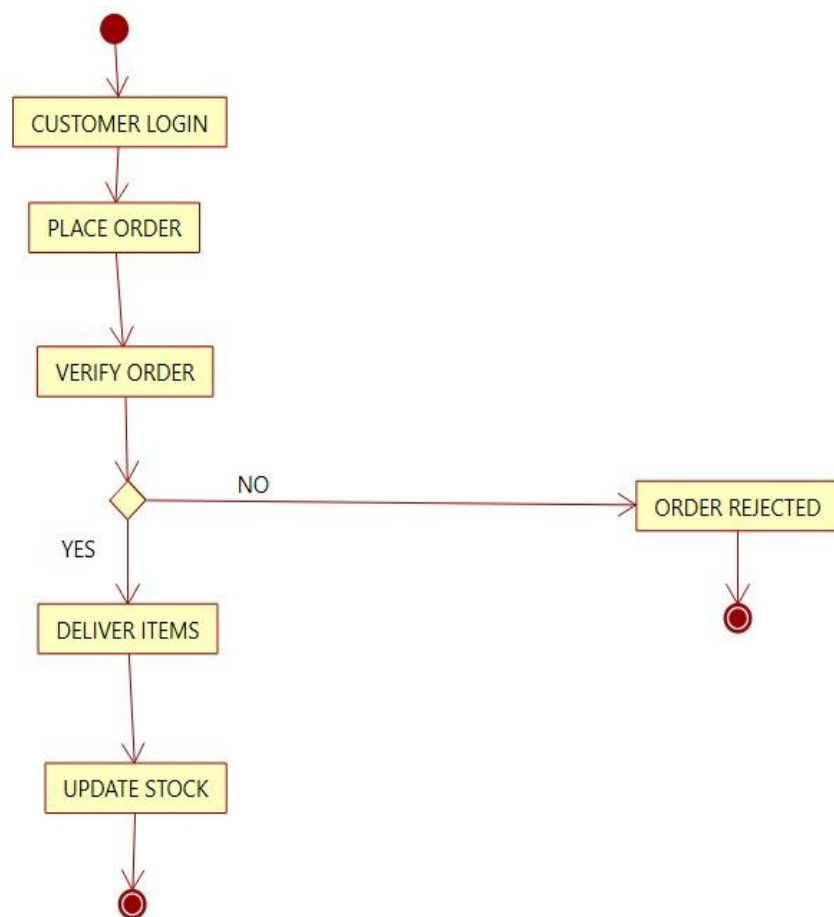
Class Diagram



Sequence Diagram



Activity Diagram



Result:

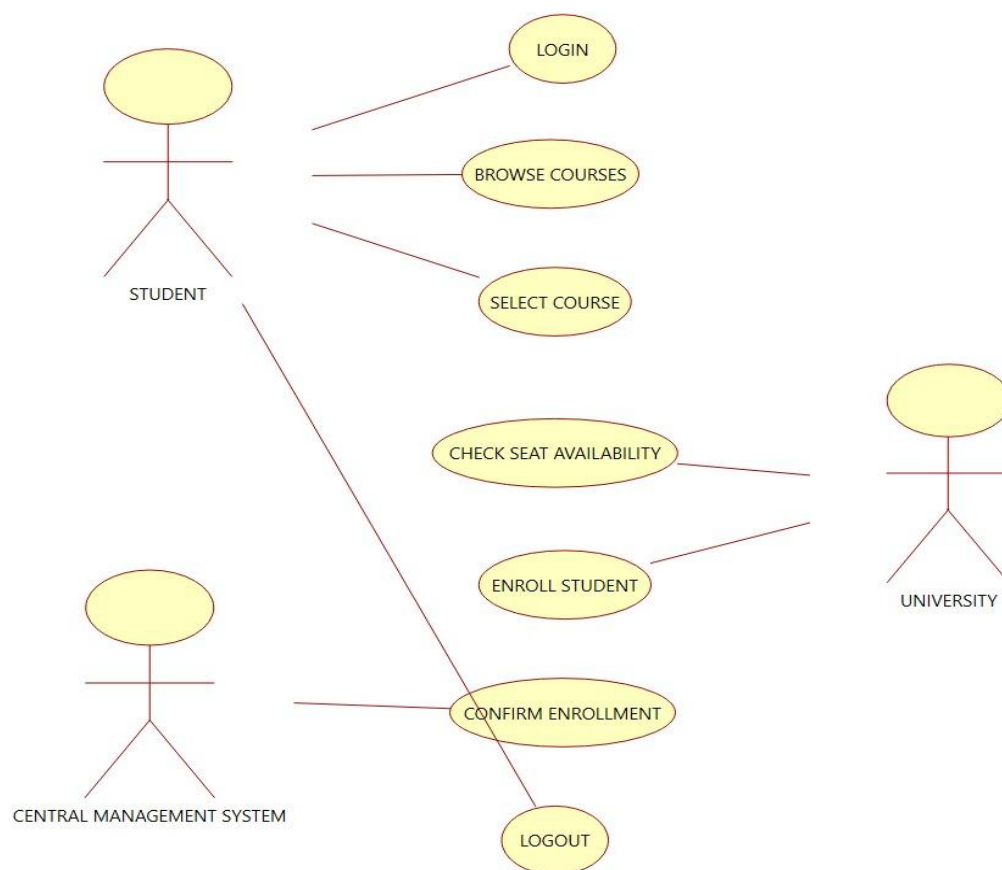
The Stock Maintenance System was successfully modeled using Umbrello UML Modeller. The UML diagrams represent order placement, verification, stock checking, item delivery, and inventory updates.

Analysis:

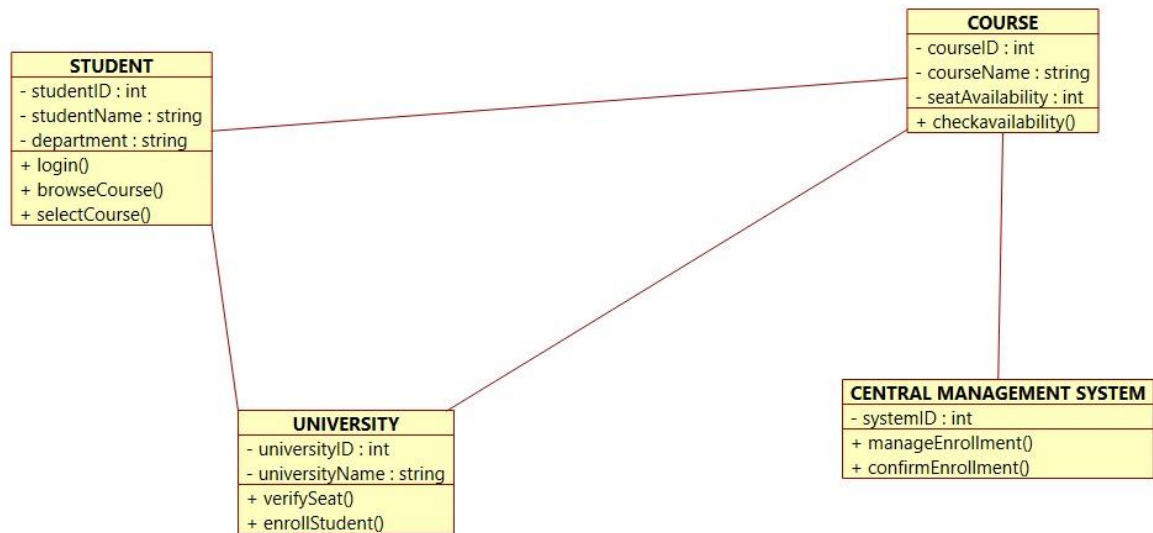
The UML model clearly illustrates the interactions among the Customer, Stock Dealer, and Central Stock System. It effectively represents the complete workflow from order placement to stock updates, satisfying the object-oriented analysis and design requirements for the experiment.

QUESTION 4: Online Course Reservation System**Aim:**

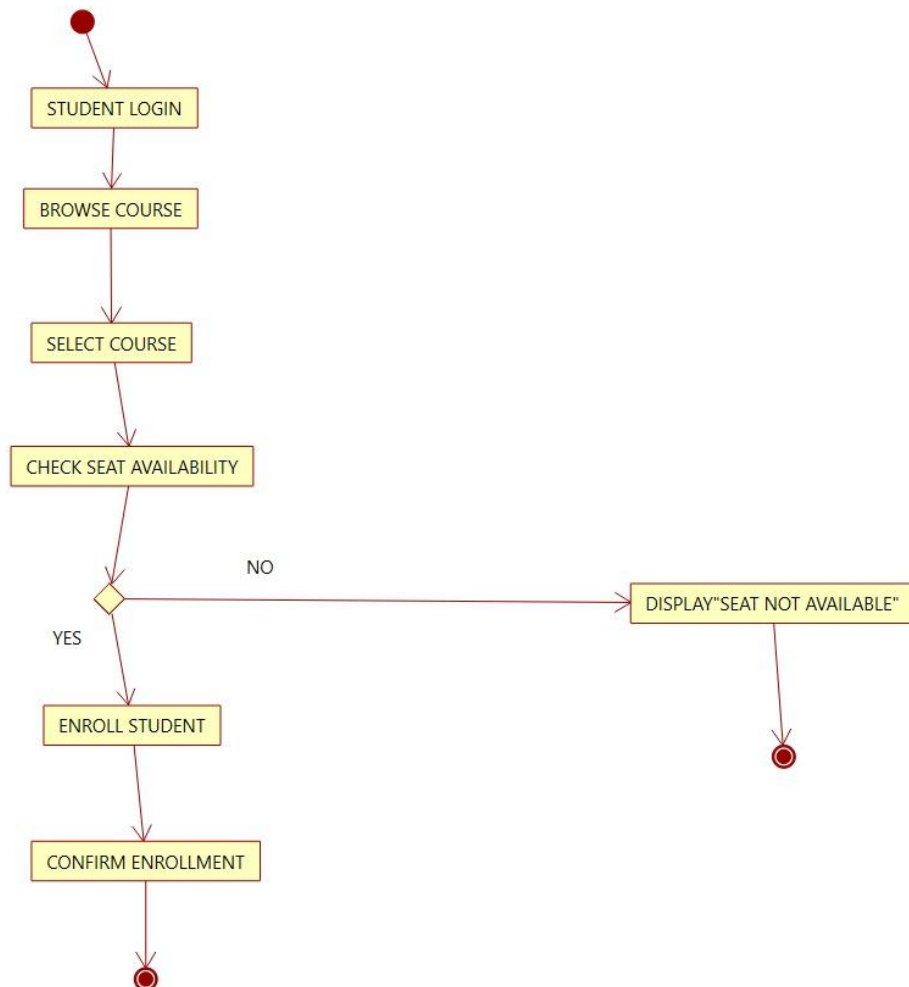
To develop a UML model for the Online Course Reservation System using Umbrello UML Modeller to represent course selection, seat availability checking, and student enrollment.

Procedure:**Use Case Diagram**

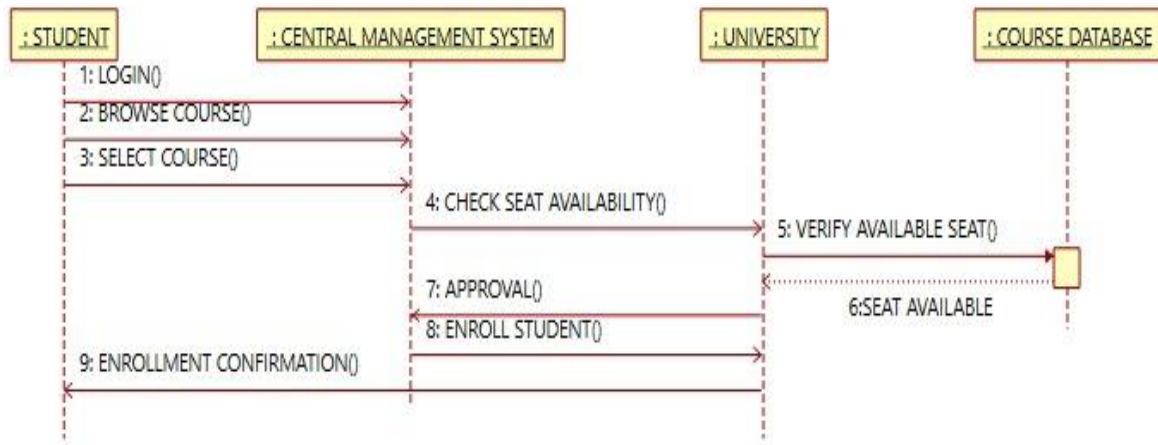
Class Diagram



Activity Diagram



Sequence Diagram



Result:

The Online Course Reservation System was successfully modeled using Umbrello UML Modeller. The UML diagrams represent course browsing, course selection, seat verification, student enrollment, and enrollment confirmation.

Analysis:

The UML model clearly illustrates the interaction among the Student, University, and Central Management System. It represents the complete workflow from course selection to enrollment confirmation, fulfilling the object-oriented analysis and design requirements specified in the experiment.